

Towards Increased Patient Engagement through Visual Support for Qualitative Questionnaire Completion and Summarization

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Motivation

- **self-reported patient data** can play an integral role in prospective clinical studies
- **low adherence to questionnaire prompts** reduces the quality of the data and therefore the research outcomes
- **visual aid** for questionnaire completion and summarization is mainly **limited** to progress bars and line and graph charts (i.e. "forgettable visualizations")

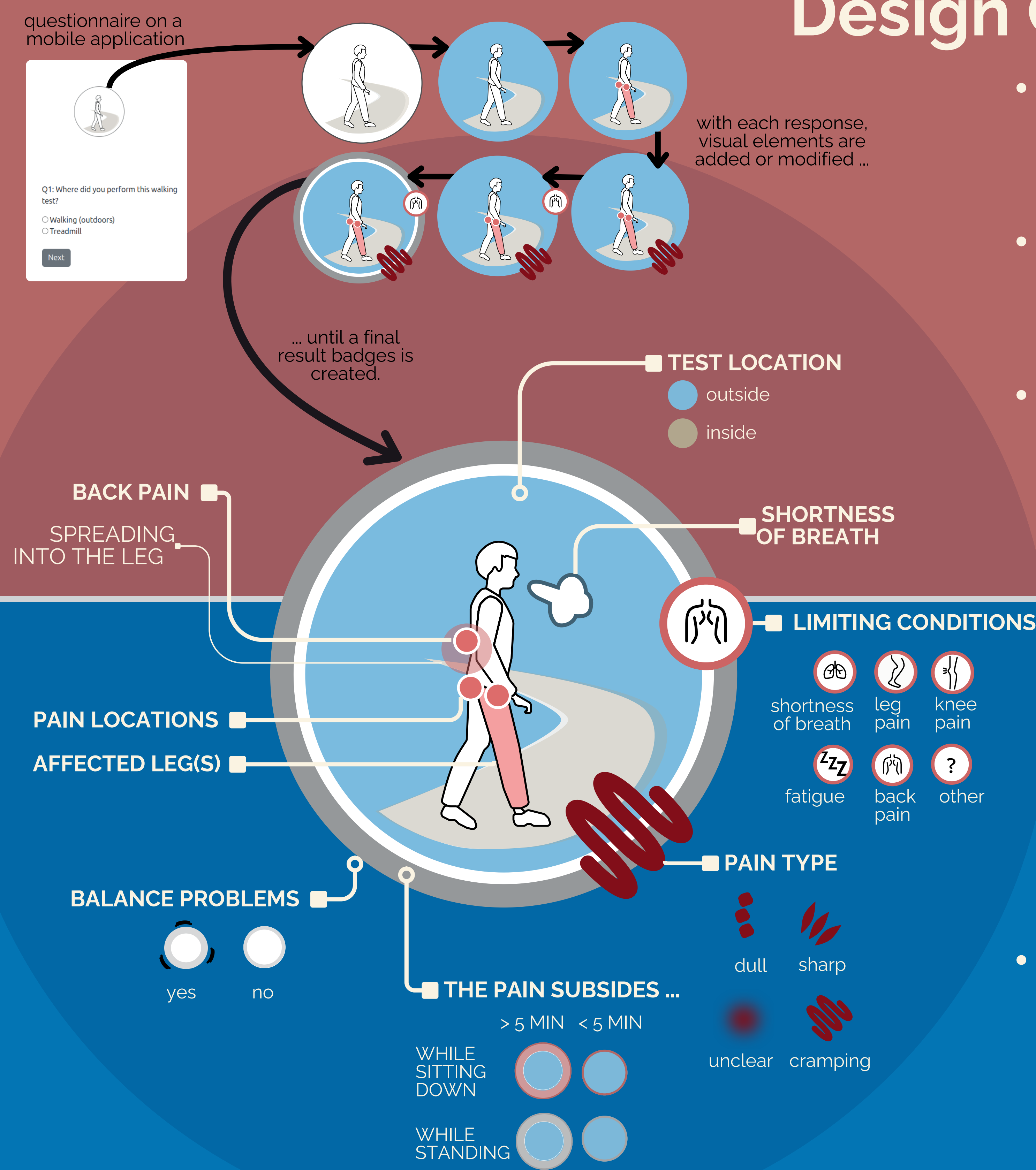
Design Considerations

- **visual analogies** outperform text and graph visualizations when it comes to understanding self-monitored health data [1]
- visual aid **during questionnaire completion** can improve question understanding and symptom perception [2]
- elements from gamification (such as **badges**) can have a positive impact for health and wellbeing related interventions [3].

Method

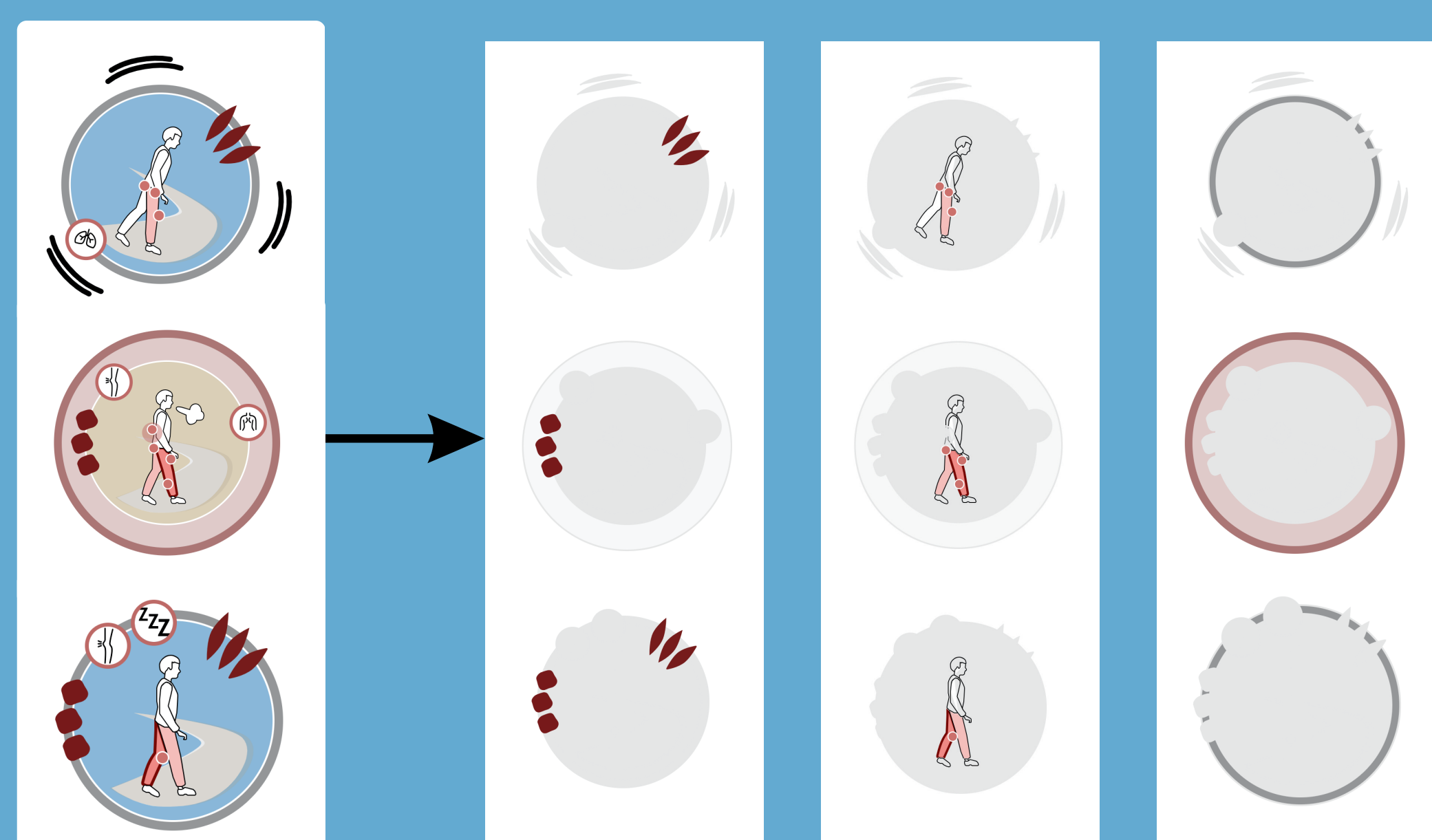
We propose a dynamic visual system for the 6MWT questionnaire (see right side), that:

- builds up an illustration step-by-step through applying coherent **metaphors** as the patient responds
- leads to a **completion badge** that visually summarizes the response set and is unique to the given answers
- reuses visual elements in a **pictograph** to track statistics over time
- allows **filtering** on the visual elements of the badges to focus on individual symptoms

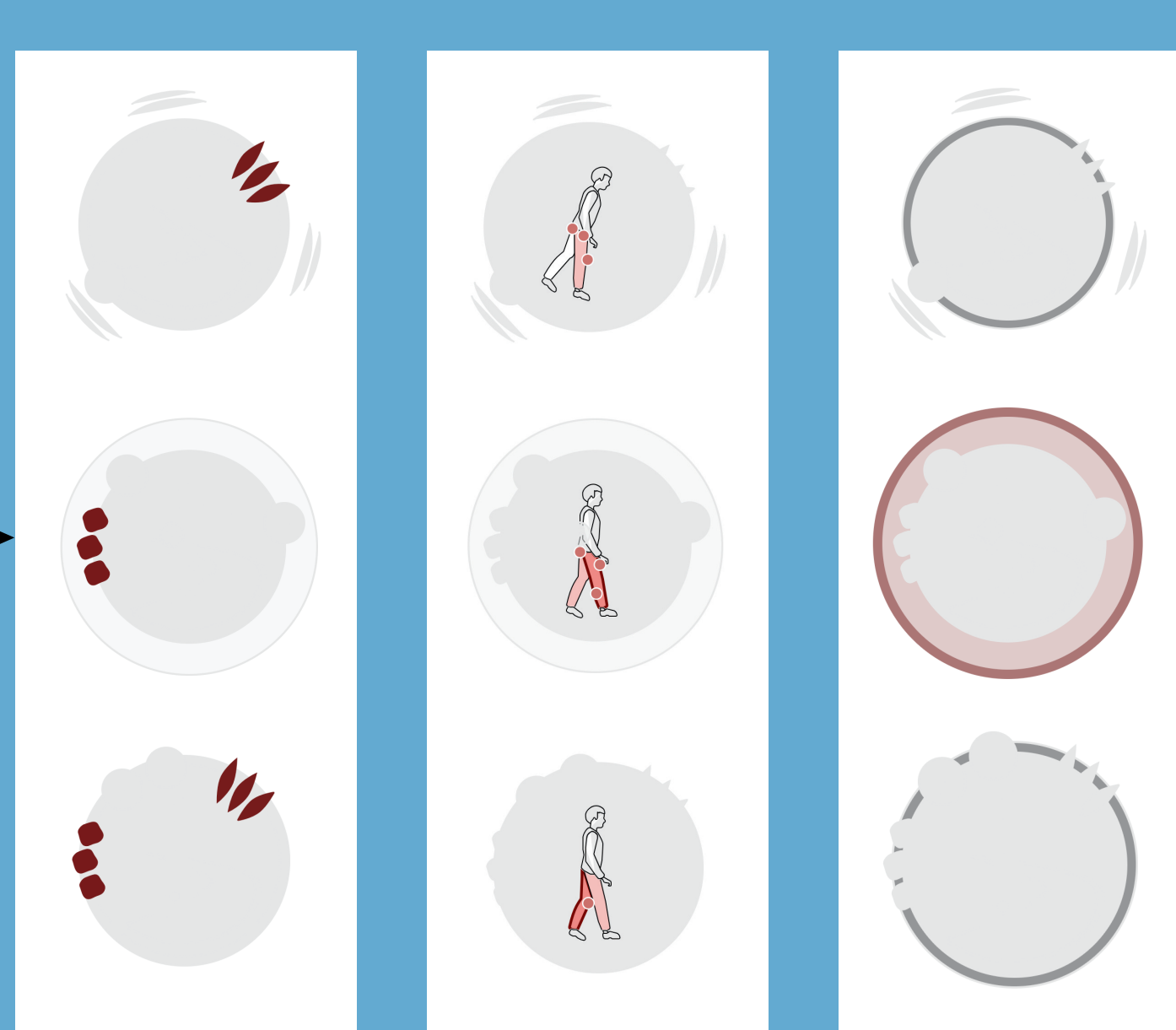


The 6-minute Walk Test (6MWT)

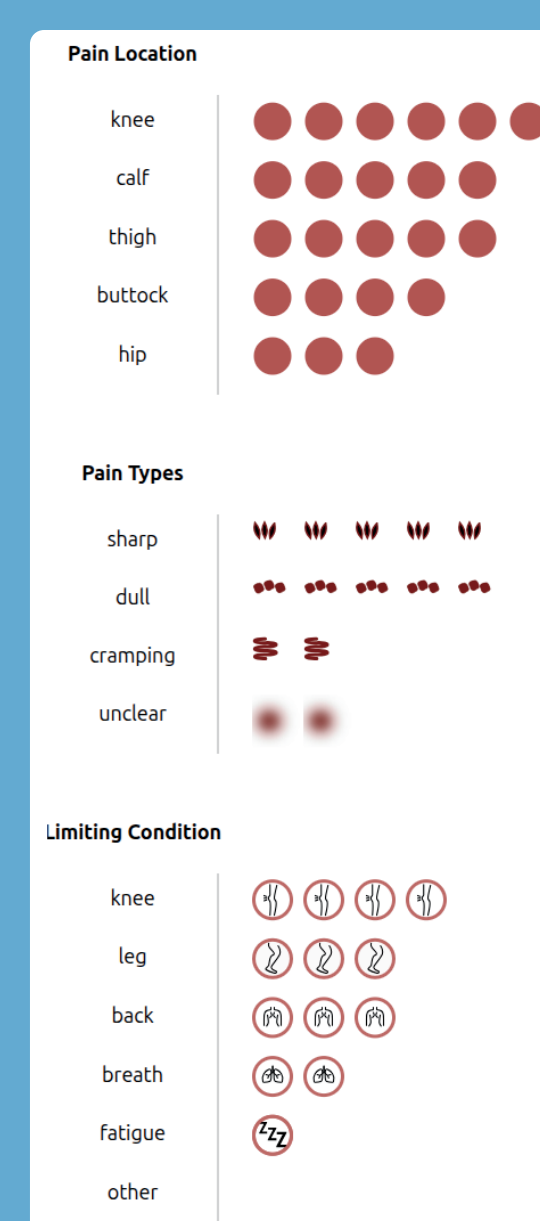
- a self-paced, low-exercises test and standardized **tool to measure clinically meaningful outcomes** across a variety of diseases, including **cardio-vascular and respiratory diseases**
- the accompanying questionnaire collects data on the patients' **symptom experience**



completion badges for three different response sets



filters on pain type (left), pain locations (middle) and subsiding of the pain (right)



pictograph for a subset of symptoms

Limitations and Outlook

Visualizing personal health data faces a plethora of challenges and has been described as "wicked problem"[4]. The prototype we are proposing only addresses a very specific challenge within this huge problem space. We built on related findings and therefore hypothesize that our approach can contribute to:

- counteracting missing data and **improve data quality** of the collected data
- maintaining **patient engagement** and adherence to questionnaire protocol

We plan to refine the proposed design and to evaluate our hypotheses in a qualitative study with a **patient focus group**.



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Acknowledgement: This research was funded by the European Union Horizon Europe Health program (HORIZON-HLTH-2022-STAYHLTH-01-two-stage) under the VASCUL-AID project (grant agreement ID: 101080947)



[1] Reading Turchioe, Meghan, et al. "Visual analogies, not graphs, increase patients' comprehension of changes in their health status." Journal of the American Medical Informatics Association 27.5 (2020): 677-689.
[2] Grossman, Lisa V., et al. "Leveraging patient-reported outcomes using data visualization." Applied clinical informatics 9.03 (2018): 565-575.
[3] Johnson, Daniel, et al. "Gamification for health and wellbeing: A systematic review of the literature." Internet interventions 6 (2016): 89-106.
[4] Rajabiyazdi, Fateme, et al. "Communicating patient health data: A wicked problem." IEEE Computer Graphics and Applications 41.6 (2021): 179-186.